

Evaluation of CLIP-Conditioned Variational Autoencoders for Image Generation Across Different Datasets

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Abstract. This paper explores the quality and effectiveness of using CLIP image embeddings as conditional inputs for a Conditional Variational Autoencoder. Conditional VAEs are widely used for image generating tasks due to their ability to adapt to different data distributions while preserving a good diversity in generation results. In this paper I am evaluating that simple yet novel approach of extracting image embeddings from the CLIP model and using them to condition a VAE for image reconstruction and generation. The evaluation is done on multiple datasets of different complexity: CIFAR-10, CelebA etc. I will discuss the limitations of that approach and compare it to state-of-the-art Text-to-Image generation models. This work contributes to research of lightweight models utilization as possible alternative to computationally expensive diffusion models.

Keywords: Variational Autoencoder · CLIP · Image Generation.

1 Motivation

Currently with the rise of Artificial Intelligence usage we see a trend on rising computational costs for state-of-the-art algorithms [1,2] for Image generation. Such algorithms like Stable Diffusion [3], DALL-E [4] and others take thousands of hours of training on large quantities of cutting-edge hardware. The inference of these models also require them being deployed on large servers and makes it inaccessible for usage on small-scale environments.

These factors add importance for research in the field of small-scale image generation algorithms which can be used on regular servers without a requirement for hundreds or thousands of state-of-the-art GPUs to be installed. These methods will allow small businesses and researchers use image generation technology without burdening costs on infrastructure. Also usage of light-weight models will reduce carbon impact of AI usage and shorten the need of constantly growing power demands of the industry.

2 Current methods and approaches

Currently there are several main approaches for increasing the effectiveness of generation models:

1. Optimization of hardware for efficient usage of deep learning models e.g. development of new chip architectures that are specialized for matrix computations as well as incorporate some model architectures on chip level.
2. Improvements of optimization algorithms. The research in that field concentrates on improvements of optimization methods for model training.
3. Cost function improvement. This field of research aims to optimize the usage of cost functions for specific tasks. These optimized cost functions allow faster convergence during training process and better results for a given task as they are designed specifically to incorporate domain specific features. One of recent advances in that field is usage of shifted-log instead of cross-entropy [11].
4. Improvements of user interaction and prompt engineering in order to generate accurate results that satisfy the user in less number of tries, which reduces the user load on servers.
5. Improvements of architecture efficiency. These methods include widely-used pruning method for removing unimportant weights or neurons from the neural network which reduces the size of the model as well as the number of FLOPS it takes to run the model. Pruning can be used for diffusion models as it is shown in paper [7]. Also there are quantization methods for large models which reduce the numerical precision of operations performed inside the model. That method heavily reduces the size of the model and makes the computations perform much faster than the original model. Model quantization is used for inference of large computationally expensive models on mobile devices - smartphones or micro controllers. Quantization of diffusion models is shown in the following paper [8].
6. Development of new algorithms for image generation that require less computational power or Improvements of smaller models for image generation to match the quality of state-of-the-art ones. That field includes Unified GAN Compression [9] and Geodesic Diffusion Models [10]

In this paper I am focusing on the last point - trying to evaluate a lightweight approach for text-to-image generation using CLIP-based text embeddings [5] as a context for a conditional VAE [6].

3 Open problems and challenges

Despite the fact that the approach of using VAE in combination with CLIP conditioning is a promising approach it still has some open challenges to be solved in order to overcome the architecture limitations. These limitations include:

1. Variational autoencoders tend to generate blurred images due to their stochastic nature of latent space. That problem can be solved via image post-processing or combining VAE with other techniques like GAN.
2. Posterior collapse of VAE. That issue represents the condition when the decoder completely ignores the latent variables and relies only on the condition. That issue is widely described in various papers and some approaches to solve it were proposed. [12,13]

3. Lack of detail in CLIP embeddings. These embeddings contain semantic information and lack in understanding finer details of the image. In combination with VAE blurred outputs it results in complete loss of details on the generated images.
4. CLIP was trained on various images from the internet and evaluating it on some datasets that have narrow domain may lead to biased results. For example CelebA that consists only of face images might result in much worse results rather than CIFAR-100 or ImageNet.

4 Contents of the paper

The paper evaluates a general overview on CLIP-conditioned Variational Autoencoder. The concept of a conditional autoencoder was introduced by Kihyuk Sohn, Honglak Lee, Xinchen Yan in their 2015 paper [15]. They used it for generation of low resolution images and object segmentation. We can apply their approach of providing a context for encoder and decoder in order to generate required output. Another algorithm that is used is CLIP [5] - a deep learning model capable of zero-shot transfer between text and images. It calculates embeddings for a text prompt and for an image and matches the ones that represent a coherent pair. In my system I calculate CLIP embedding for the image and then pass it as a condition for the encoder and decoder parts of the variational autoencoder. That allows the model to capture the context of the image and because text and image embeddings are matched I can use text prompts to condition the decoder for image generation. A similar approach but for GANs was proposed in the papers [14] and [16] but their approach differs from the one proposed in that paper. The evaluation of that approach is done on different datasets such as CelebA, Cifar-10 and others. The results of such evaluation can be used as a reference point for future work in the field of CLIP and VAE applications for image generation.

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